

P4.10

First we'll check the total mass balance:

$$\text{IN} = 134 \text{ g} + 134 \text{ g} = 268 \text{ g}$$

$$\text{OUT} = 27 \text{ g} + 52 \text{ g} + 121 \text{ g} + 68 \text{ g} = 268 \text{ g}$$

So total mass is OK.

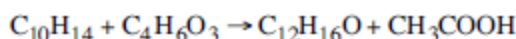
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We can proceed by either (1) checking element mass balances or (2) converting to moles, and checking stoichiometry. We will take approach 2 here.

$$\text{IN: } 134 \text{ g IBB} \times \frac{1 \text{ gmol IBB}}{134 \text{ g IBB}} = 1 \text{ gmol IBB}, 134 \text{ g AAn} \times \frac{1 \text{ gmol AAn}}{102 \text{ g AAn}} = 1.314 \text{ gmol AAn}$$

$$\text{OUT: } 27 \text{ g IBB} \times \frac{1 \text{ gmol IBB}}{134 \text{ g IBB}} = 0.20 \text{ gmol IBB}$$

Therefore, 0.80 gmol IBB are consumed by reaction. The balanced chemical reaction is



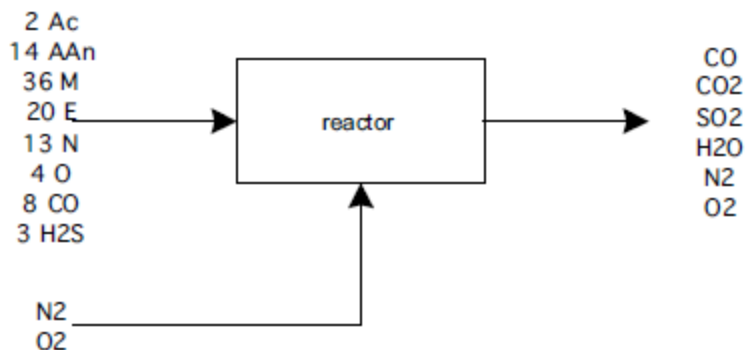
Balances on other compounds:

	In (moles)	Consumed or generated (moles)	Out (moles)	Molar mass	Out (grams)
$\text{C}_4\text{H}_6\text{O}_3$	1.314	-0.8	0.514	102	52.4
$\text{C}_{12}\text{H}_{16}\text{O}$	0	+0.8	+0.8	176	140.8
CH_3COOH	0	+0.8	+0.8	60	48

The last column is calculated from material balances. Agreement with the data for acetic anhydride is good, but the calculated masses of IBA and acetic acid are different from the data by 20 g. Thus, the chemist's data does not satisfy material balance requirements, and must be incorrect.

P4.19

Assuming a convenient basis of 100 gmol/min fed, the flow diagram becomes:



There are 13 compounds but only 5 elements, so an element balance approach is attractive.

C balance:

$$2(3) + 14(4) + 36(1) + 20(2) + 8(1) = 146 = \dot{n}_{CO_2, out} + \dot{n}_{CO, out}$$

H balance:

$$2(6) + 14(6) + 36(4) + 20(6) + 3(2) = 366 = 2\dot{n}_{H_2O, out}$$

$$\dot{n}_{H_2O, out} = 183$$

S balance:

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$$3(1) = 3 = \dot{n}_{SO_2, out}$$

$$\dot{n}_{SO_2, out} = 3$$

O balance:

$$2(1) + 14(3) + 4(2) + 8(1) + 2\dot{n}_{O_2, in} = 2\dot{n}_{O_2, out} + 183 + 3(2) + \dot{n}_{CO, out} + 2\dot{n}_{CO_2, out}$$

$$2\dot{n}_{O_2, in} = 2\dot{n}_{O_2, out} + \dot{n}_{CO, out} + 2\dot{n}_{CO_2, out} + 129$$

N balance:

$$13(2) + 2\dot{n}_{N_2, in} = 2\dot{n}_{N_2, out}$$

$$2\dot{n}_{N_2, in} = 2\dot{n}_{N_2, out} - 26$$

We still have 6 unknowns, but we have not yet used all our specifications. First, we know the composition of the inlet air, or

$$\frac{\dot{n}_{O_2,in}}{\dot{n}_{N_2,in}} = \frac{21}{79}$$

We also know that oxygen is fed at 20% excess, calculated on the assumption of complete combustion. We can use our element balance equations to first solve the case of complete combustion with a stoichiometric oxygen feed rate, where there is no CO and no O_2 in the effluent. This allows us to find the oxygen (and nitrogen) feed rate at 0% excess, thereby allowing us to calculate the oxygen feed rate at 20% excess. Then we use the calculated air feed rate and the specification on the actual CO:CO₂ ratio to find the actual burner outlet.

Complete combustion case: solve equations above for $\dot{n}_{CO,out} = 0$ and $\dot{n}_{O_2,out} = 0$:

$$\begin{aligned}\dot{n}_{CO_2,out} &= 146 \\ \dot{n}_{O_2,in} &= \frac{2(146) + 129}{2} = 210.5 \\ \dot{n}_{N_2,in} &= \frac{79}{21}(210.5) = 792\end{aligned}$$

Actual case: At 20% excess, the oxygen in is 1.2(210.5) or 252.6 gmol/min (at the chosen basis) and the nitrogen in is 1.2(792)=950.4.

Also, we know that $\frac{\dot{m}_{CO_2,out}}{\dot{m}_{CO,out}} = \frac{4}{1}$. This is a mass ratio, but we are working in moles, so we need the molar masses to convert to a mole ratio:

$$\frac{\dot{n}_{CO_2,out}}{\dot{n}_{CO,out}} = \frac{\dot{m}_{CO_2,out}}{\dot{m}_{CO,out}} \frac{M_{CO}}{M_{CO_2}} = \frac{4}{1} \left(\frac{28}{44} \right) = 2.55$$

Going back to our element balance equations with these changes and solving, we find:

$$\begin{aligned}\dot{n}_{N_2,out} &= 963 \\ \dot{n}_{CO_2,out} &= 104.9 \\ \dot{n}_{CO,out} &= 41.4 \\ \dot{n}_{O_2,out} &= 62.65\end{aligned}$$

The total flow out of the burner is 183+3+963+104.9+41.4+62.65=1358, so the burner outlet composition is: 4.6 mol% O₂, 70.9 mol% N₂, 3.0 mol% CO, 7.7 mol% CO₂, 13.5 mol% H₂O, 0.22 mol% SO₂.